

Date

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Anova Test

→ Analysis of Variance (ANOVA) : It's a statistical method used to compare means of 2 or more groups.

⇒

① Factors (variable)

② Levels

⇒ e.g., Medicine (factor)
[dosage] 5mg 10mg 15mg (levels)

Assumptions in Anova:

⇒ ① Normality of sampling distⁿ of Mean
The distⁿ of sample mean is normally distributed.

② Absence of Outliers

Outlying score need to be removed

③ Homogeneity of variance

Popⁿ variance in diff. levels of each ind. variable are equal. $[\sigma_1^2 = \sigma_2^2 = \sigma_3^2]$

(इस से है)

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④ Sample are ind. & randomly selected

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Types of Anova:

① One way Anova: one factor with atleast 2 levels, these levels are independent.

② Two way Anova:

③ Factorial Anova:

Hypothesis testing in ANOVA:

$\Rightarrow$ Null Hyp. $H_0: \mu_1 = \mu_2 = \dots = \mu_k$

Alternate Hyp. H_1 : atleast one of the sample means is not equal

F-test:

$$F = \frac{\text{Variance b/w samples}}{\text{Variance within sample}}$$

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